Customer Churn Classification – auto model training

1. Introduction

Customer churn prediction identifies likely departing customers to enable timely retention strategies. Accurate prediction is critical for reducing business losses and improving customer lifecycle management.

2. Dataset Overview

The dataset contains demographic, service usage, and billing information with a binary churn target. Key points:

- **Rows/Columns:** Several thousand records, multiple features
- **Feature Types:** Numerical (tenure, monthly charges), Categorical (contract type, payment method)
- **Preprocessing:** Minimal missing values, handled with imputation; categorical features encoded; numerical features scaled.

3. Preprocessing & Augmentation

A reusable pipeline was applied for all models:

- Missing value handling
- Categorical encoding
- Numerical scaling
- Train/test split (80/20)

Data Augmentation: Bootstrapping and noise injection applied to training set for retraining, processed through the same pipeline to maintain feature consistency.

4. Models Implemented

- **Logistic Regression (Scratch):** Gradient descent implementation; validates core optimization logic.
- **Logistic Regression (Sklearn):** Linear model with regularization.
- **KNN:** Instance-based, proximity-driven.
- **SVM:** Margin-based, high-dimensional suitability.
- **Decision Tree:** Non-linear, interpretable rules.
- **Random Forest:** Ensemble of trees, improves generalization and stability.

5. Training Setup

- Train/Test Split: 80/20
- Metrics: Train Accuracy, Test Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix
- Hyperparameters: Default settings, focus on retraining effect and model behavior

6. Results & Comparison

6.1 Initial Training

Model	Train Acc	Test Acc	Precision	Recall	F1-score	ROC-AUC	Confusion Matrix
Decision Tree	0.8085	0.7928	0.6385	0.5053	0.5642	0.8364	[[928,107],[185,189]]
KNN	0.8080	0.7814	0.5902	0.5775	0.5838	0.8298	[[885,150],[158,216]]
Logistic Regression (Scratch)	0.7623	0.7637	0.5411	0.7219	0.6186	0.8390	[[806,229],[104,270]]
Logistic Regression (Sklearn)	0.7987	0.7828	0.5853	0.6257	0.6047	0.8418	[[869,166],[140,234]]
Random Forest	0.7936	0.7580	0.5298	0.7834	0.6321	0.8419	[[775,260],[81,293]]
SVM	0.7725	0.7438	0.5115	0.7727	0.6155	0.8211	[[759,276],[85,289]]

6.2 Retraining with Augmentation

Model	Train Acc	Test Acc	Precision	Recall	F1-score	ROC-AUC	Confusion Matrix
Decision Tree	0.8156	0.7949	0.6641	0.4599	0.5434	0.8195	[[948,87],[202,172]]
KNN	0.7970	0.7814	0.6071	0.5000	0.5484	0.8177	[[914,121],[187,187]]
Logistic Regression (Scratch)	0.7143	0.7353	0.5009	0.7674	0.6061	0.8318	[[749,286],[87,287]]
Logistic Regression (Sklearn)	0.7814	0.7949	0.6375	0.5267	0.5769	0.8309	[[923,112],[177,197]]
Random Forest	0.8464	0.7445	0.5119	0.8021	0.6250	0.8419	[[749,286],[74,300]]
SVM	0.7833	0.7480	0.5172	0.7647	0.6170	0.8163	[[768,267],[88,286]]

Observations:

- Retraining amplifies inductive bias: recall-heavy models (Random Forest, SVM) see improved recall; precision-focused models (Decision Tree) slightly degrade in precision.
- Tree ensembles gain stability in recall but may overfit to augmented data.
- Linear models maintain consistent performance, confirming preprocessing and implementation correctness.
- Scratch Logistic Regression aligns closely with sklearn, validating custom code.

7. Automated Retraining Analysis

The retraining pipeline combines original and augmented data for model refitting. Metrics and confusion matrices for retraining rounds are recorded systematically for analysis.

Insights:

- Recall gains are model-dependent; precision can be negatively affected for some architectures.
- Model ranking remains consistent across rounds: Random Forest > Logistic Regression (Sklearn) > Decision Tree/KNN/SVM.
- Low metric variance indicates robust pipeline implementation and reproducibility.

8. Scratch Logistic Regression Validation

Custom implementation correctly converges using gradient descent. Metrics closely match sklearn, confirming algorithmic correctness without relying on library optimizations.

9. Conclusion

- **Top Performer:** Random Forest for recall, capturing churners effectively.
- **Balanced Performer:** Logistic Regression (Sklearn) for stable metrics.
- **Retraining Insights:** Enhances recall for under-detecting models; preserves linear model stability.
- **Technical Takeaway:** The automated retraining pipeline reliably evaluates and updates models with augmented data, demonstrating reproducible behavior across multiple training rounds.